

Qatar-2b

R-1136

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_210222 · created 2026-07-30T23:35:42+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459267.91156 +/- 0.00093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.015
Transit depth (Rp/Rs)^2	3.3 +/- 0.56 [%]
Orbital Inclination (inc)	86.73 +/- 1.57
Airmass coefficient 1 (a1)	1.019 +/- 0.019
Airmass coefficient 2 (a2)	-0.011 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#3 - [602, 218]
Optimal Method	PSF photometry
Transit Duration (day)	0.0735 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.182 ± 0.015 vs 0.16327 (deviation 1.25σ)
Duration fit vs published	0.0735 d vs 0.0767 d (deviation 0.7σ)
Mid-time O-C	2.7 min
Depth SNR	12.1
Residual scatter	1.89 %

Light curve

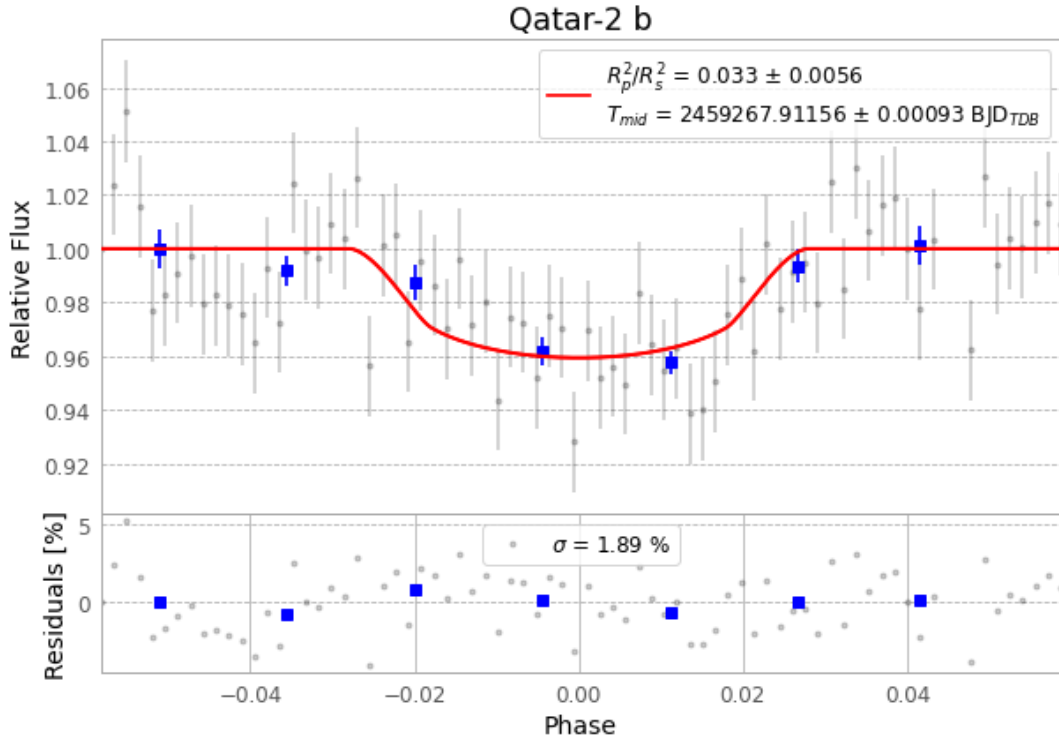


Figure 1 — FinalLightCurve_Qatar-2 b_2021-02-22.png (SHA-256 9158266d63842b32...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [573, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eb72b14b682993da...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), Qatar-2210222075412.FITS → Qatar-2210222113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-210222.log (fa761feaacb4...), FinalLightCurve_Qatar-2 b_2021-02-22.png (9158266d6384...), FinalLightCurve_Qatar-2 b_2021-02-22.csv (e946a41d893d...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 23:35Z.